

Residual UMAP of prediction change $\Delta\hat{y} = \hat{y}^{(1)} - \hat{y}^{(0)}$

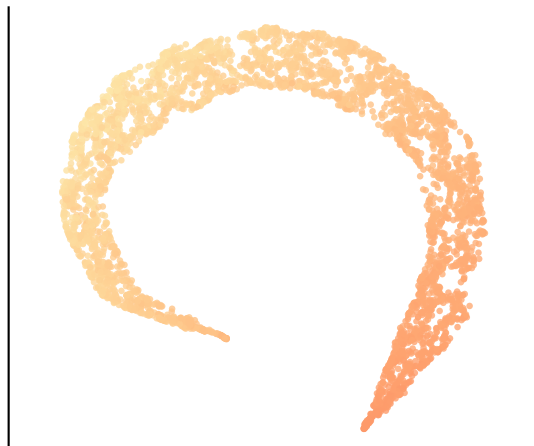
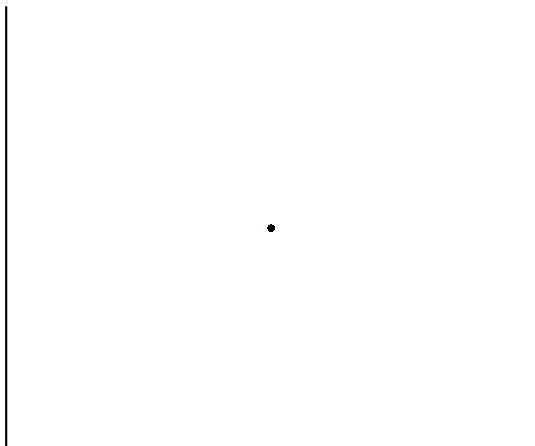
Math1 random split — change-only embedding (amplifies ablation contrast)

CLEAN-Full

CLEAN (w/o OrthoMask)

UMAP($\Delta\hat{y}$), mean $\|\Delta\hat{y}\|_2=0.000$

UMAP($\Delta\hat{y}$), mean $\|\Delta\hat{y}\|_2=0.778$



$\|\Delta\hat{y}\|_2$

0.8

0.7

0.6

0.5

0.4

0.3

0.2

0.1

0.0

CLEAN (w/o FrozenBias)

UMAP($\Delta\hat{y}$), mean $\|\Delta\hat{y}\|_2=0.165$



CLEAN (w/o OCM)

UMAP($\Delta\hat{y}$), mean $\|\Delta\hat{y}\|_2=0.755$

